(A)

Bangladesh Army University of Engineering & Technology (BAUET)

Department: CSE, Batch: 18th, Course no: CSE-1213, CT-1, Date: 19-9-2024

1. Write the logical expression for the sentence "If you get 80% on the final examination, then you will get an A+" and also the truth table with all possible input combinations. 5
2. Write down the truth table for the $p \rightarrow q$ and $p \leftrightarrow q$ propositions. 5
3. Construct the truth table of the compound proposition $(p \vee \neg q) \rightarrow (p \wedge q)$. 5

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Bangladesh Army University of Engineering & Technology
Department of Computer Science & Engineering
CT: 2, Time: 30 mins

...ind the truth set of each of these predicates where the domain is the set of integers.

i) $x^2 > 2$. ii) $x^2 > x$. (iii) $2x + 1 = 0$.

Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.

Show that if n is an integer and $n^3 + 5$ is odd, then n is even using a proof by contradiction.

$x = 1$ $1 \neq 2$

$x = 2$ $4 > 2$

$2k+1 \Rightarrow 2k$

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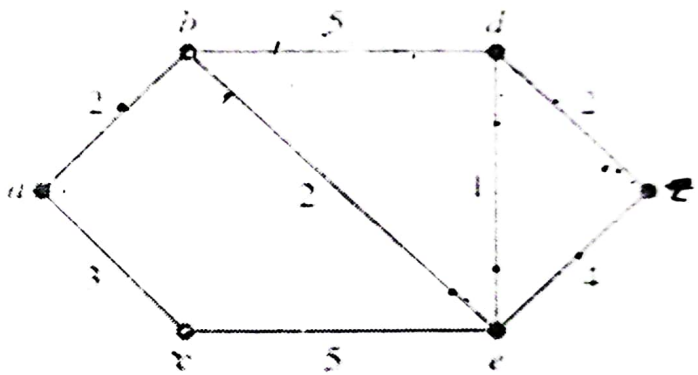
CT-03

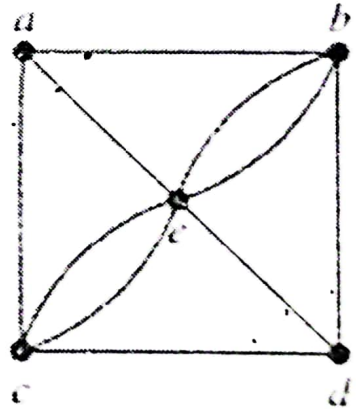
Bangladesh Army University of Engineering & Technology (BAUET)

Department: CSE, Batch: 18th, Course no: CSE-1213, CT-3, Date: 7-11-2024, Time: 20 mins

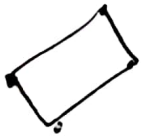
1. Define weighted graph. Find the degree and adjacency matrix of the given graph. 8



2. Define Euler path and circuit. Determine whether the given graph has Euler path and circuit. If yes, write down at least one path and a circuit. 7



vertex



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